

Functional Description

Diesel engine-generator sets

DP00275D5SFA

DP00300D5SFA

DS00300D5SFA

DS00330D5SFA

Application groups 3B, 3D

Built in Europe

MS13034/02E



System designation	Engine model	Application group
DP00275D5SFA	6R1600G10F	3B, Continuous operation, variable
DP00300D5SFA	6R1600G20F	3B, Continuous operation, variable
DS00300D5SFA	6R1600G70F	3D, Short-term operation, fuel stop
DS00330D5SFA	6R1600G80F	3D, Short-term operation, fuel stop

Table 1: Applicability

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1 Engine-Generator Set

1.1 Engine-generator set – General information

The engine-generator set consists mainly of a diesel engine and a generator mounted on a common base frame. Upon demand, the engine will start and drives the generator to produce electrical power.

Engine-generator sets are available in two application groups: prime power and standby power.

Application group 3B – Prime power

Prime power is continuous electric power generation for long periods of time. Engine-generator sets for prime power are used in the following conditions:

- No utility service available, e.g. in remote or developing areas
- Existing utility service is very expensive or not reliable

Continuous operation	Application group 3B
Operating mode	Continuous operation, variable
Rating definition	10 % overload capacity (ICXN)
Load factor	< 75 %
Operating hours	unrestricted

Application group 3D – Standby power

Standby power systems are a type of system to provide backup resources in a crisis or when regular systems (e.g. power grid) fail. They find uses in a wide variety of settings:

- Residential homes
- Hospitals
- Airports
- Scientific laboratories
- Data centers

Standby operation	Application group 3D
Operating mode	Short-time operation, variable
Rating definition	ISO fuel stop power (IFN)
Load factor	< 85 %
Operating hours	max. 500 hours per year

Model designation of engine-generator sets

Key to the engine-generator set model designation	
D	Type: Diesel engine-generator set
P / S	Application: • P – Application group 3B • S – Application group 3D
00650	Performance: Rated power in kVA
D	Fuel type: Diesel
5	Frequency: 50 Hz
S	Project type: Standard product

Benefits

- Wide range of standardized engine-generator sets to meet customer requirements with regard to power, emission and other characteristics
- Possibility to choose from different component designs (e.g. fuel prefilters) and to select options (e.g. fuel cooler)
- Latest diesel engine technology
- State-of-the-art main components for high efficiency and long service life

1.2 Engine-generator set – Scope of supply

The figures show an overview of the components on an engine-generator set with a 6 R 1600 Gx0 engine.

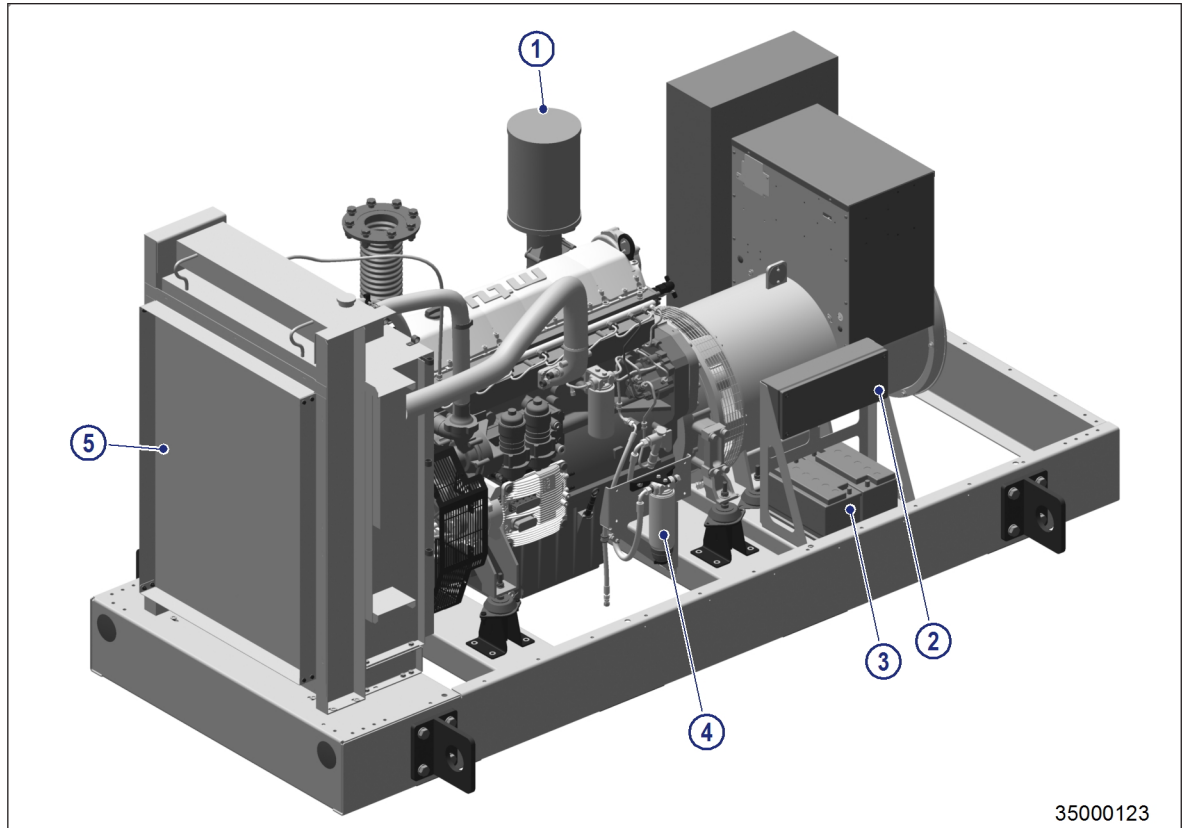
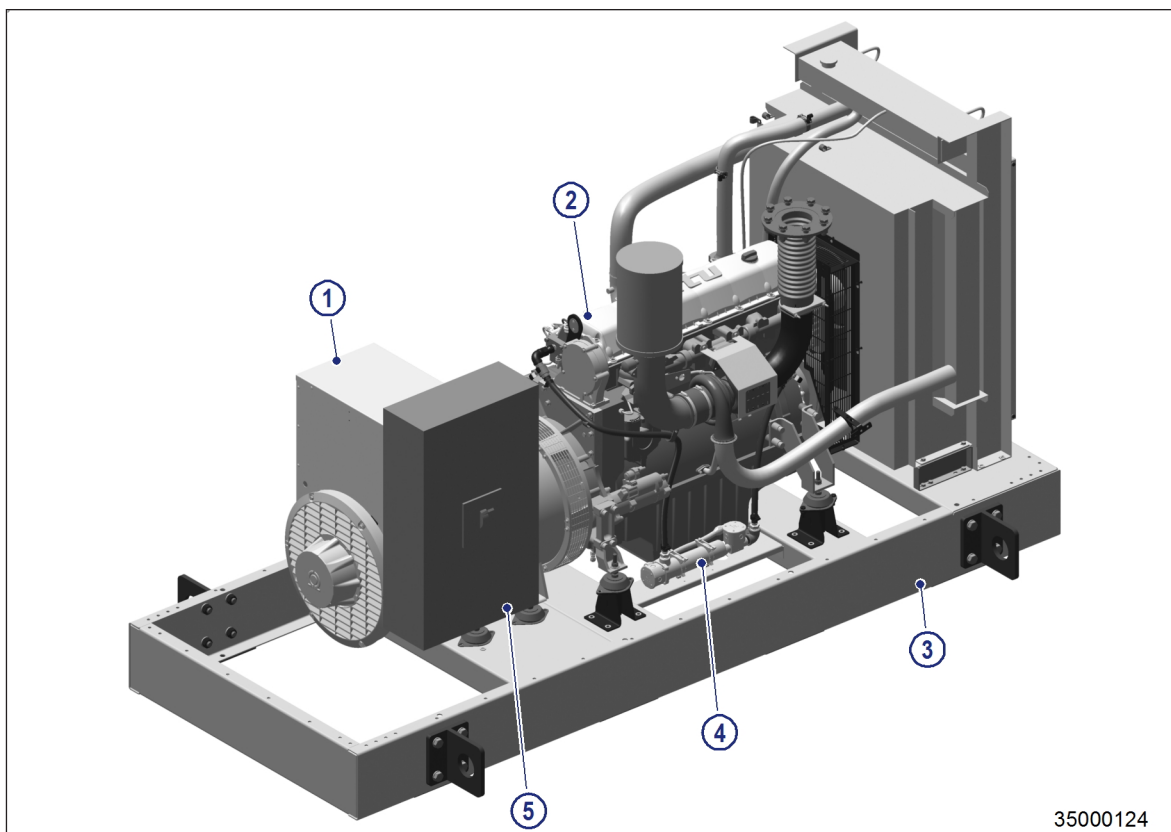


Figure 1: Engine-generator set – Coolant cooler side view

- | | | |
|------------------------------------|----------------------|------------------|
| 1 Air filter | 3 Battery (optional) | 5 Coolant cooler |
| 2 Terminal box (control version 1) | 4 Fuel prefilter | |



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Figure 2: Engine-generator set – Generator side view

1 Generator

2 Engine

3 Base frame

4 Preheater (optional)

5 Circuit breaker (optional)

Scope of supply

Standard scope of supply

- MTU Series 6 R 1600 Gx0 engine:
 - 1,500 RPM (for 50 Hz)
 - Air-to-air charge-air cooling
 - Electronic engine management system ADEC (ECU8)
- Generator:
 - Voltages 400 V
 - Analog voltage regulator
 - Three CTs for measuring
 - IP-20 protection
- Control version 1: Terminal box without control function (interface between engine-generator set and customer control)
- Smart Connect
- Coolant cooler
- Air filter
- Oil drain
- Fuel prefilter
- Guards for all rotating parts
- Base frame

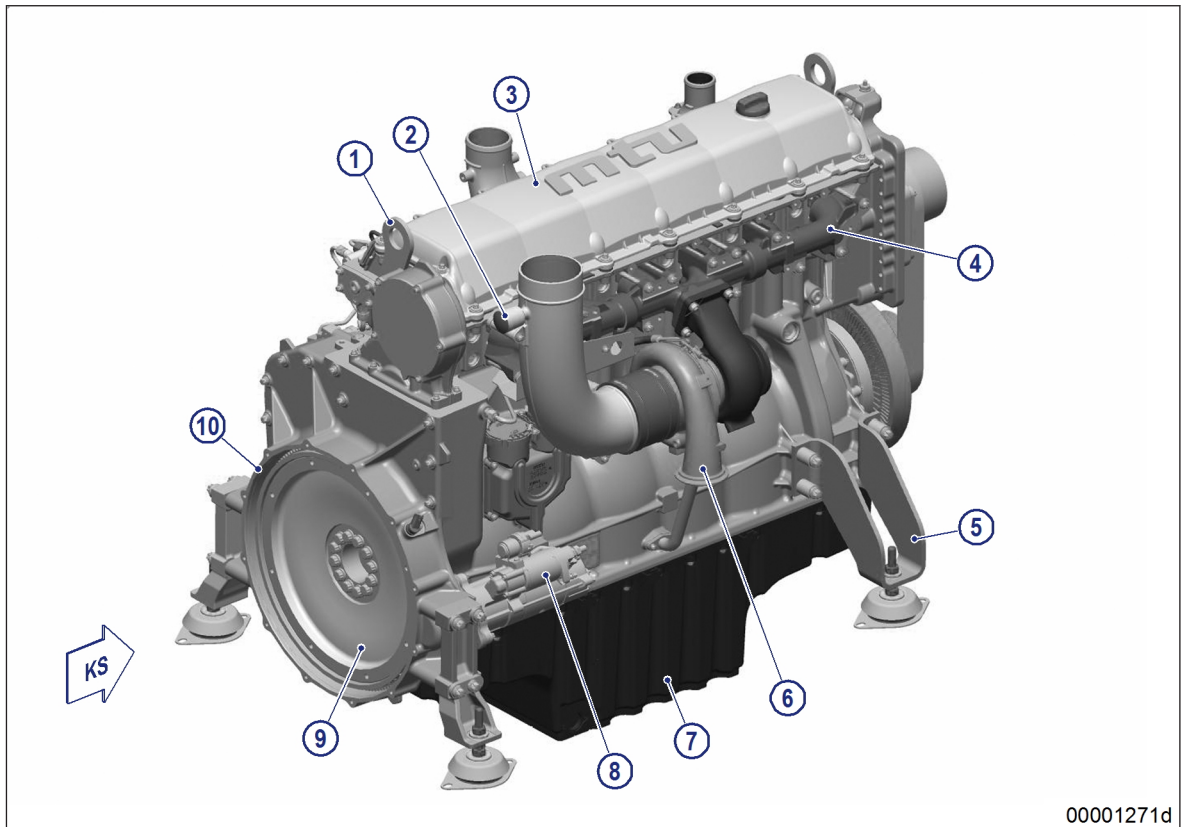
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Available options

- Generator:
 - Voltages 380 V, 415 V
 - Kit of three CTs for differential measuring
 - PMG-Kit with Digital voltage regulator
 - Anti-condensation heater
 - IP-23 drip cover
- Control version 2, 3a, 3b or 4:
 - Control panel DGC-2020 for management and monitoring
 - Control cabinet equipped with battery charger
- Fuel prefilter with water separator (single or double)
- Heavy duty air filter
- Exhaust bellows
- Exhaust elbow
- Exhaust silencer
- Heat shield
- Preheater for engine coolant
- Fuel cooler
- Fuel tank:
 - Day tank
 - Sub-base tank
- Starter batteries with battery rack and cables
- Battery charger (available as option in control version 1)
- Circuit breaker
- Enclosure

2 Engine

2.1 MTU 6 R 1600 Gx0 engine



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Figure 3: Engine view on driving end

- | | | |
|-----------------------------|------------------------|-----------------------|
| 1 Engine lifting equipment* | 5 Engine mounting | 9 Flywheel |
| 2 Service indicator | 6 Exhaust turbocharger | 10 Flywheel housing |
| 3 Cylinder head | 7 Oil pan | KS Engine driving end |
| 4 Exhaust elbow | 8 Starter | |

* Do not use these lifting points to lift the entire engine-generator set.